

ExpressiveWorld: Detachable Expressions for VR Collaboration

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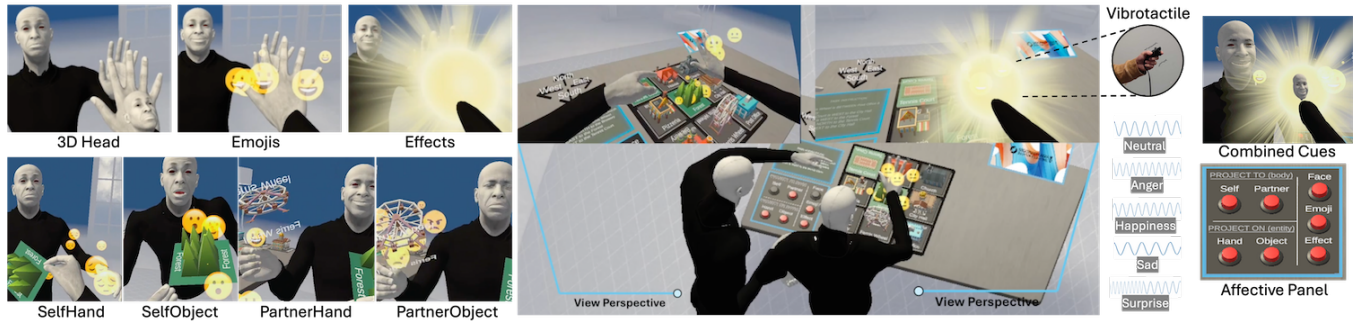


Figure 1: Our System enables VR users to collaborate while customising how their partner's emotions are seen and felt.

Abstract

VR collaboration enables users to express their emotions through visual cues. Although various techniques for representing emotions have been proposed, the impact of different representation locations on their effectiveness remains unclear. This research investigates how reprojecting a collaborator's emotions to various locations can enhance the reception of affective cues in VR collaboration. We introduce ExpressiveWorld, a customizable collaborative space that allows VR users to configure both the type and location of their partner's emotion representations. Our system provides four representations methods: 3D head models, emojis, visual effects, and vibrotactile feedback, representing five distinct emotions: Neutral, Happy, Angry, Sad, and Surprised. These representations can be reprojected onto the user's or partner's hand and task objects within the VR environment. With the ability to customize their emotional representations and positions on or off independently, users can create up to 84 different configurations. This flexibility helps VR users maintain focus on tasks while achieving emotional understanding and consensus with their partners.

Keywords

Emotion, Visual Cue, Virtual Collaboration, VR, Emoji, Effect

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1 Introduction

Virtual Reality (VR) enables users to share emotions effectively and naturally through visual and non-visual cues. Over time, novel techniques have been proposed to enhance the perception of emotions [Somarathna et al. 2023]. Common methods include sharing facial expressions, emojis, and visual effects. Each method carries unique attributes and benefits, considering users' diverse backgrounds and their accuracy in interpreting the underlying emotions. However, limited work has focused on positioning these emotional cues in a VR environment, despite its potential to enhance user experience.

We introduce ExpressiveWorld (Figure 1), an affective VR toolkit that enables users to configure the representations of their task partner's emotions while collaborating on an object interaction task. Users can selectively perceive their partner's emotions through facial expressions (3D Head), facial emojis (Emojis), visual effects (Effects), or via vibrotactile feedback that represents a heartbeat, an indicator of arousal.

These emotional representations can be displayed closer to the task area of interest to facilitate easier emotional understanding. We propose four positions: the user's hand, the user's task object, the partner's hand, and the partner's task object. The representation and position of emotions can be adjusted through an affective panel. This paper discusses the design and development of ExpressiveWorld and its applications in collaborative activities. It presents

concepts for emotion-aware collaboration to enhance emotion consensus when working together on tasks, including brainstorming sessions, problem-solving activities, design reviews, training simulations, and team-based games.

2 System Overview

Two users wear Meta Quest Pro head-mounted displays (HMDs) are immersed in a network-connected VR collaborative environment. We use the VR HMD's built-in sensors and trackers to capture users' eye gaze, facial expressions, body gestures, head positions, and head rotations. This information is used to visualize their virtual avatars, emotional visual cues (3D Head, Emojis, and Effects), generating vibrotactile feedback, and facilitate object interaction. While object interactions can be accomplished through hand gestures, our system also supports using Quest VR controllers as an alternative approach. This method provides access to vibrotactile feedback to perceive the partner's emotions in the form of heartbeats.

We utilized avatars with face tracking supported by the Meta Movement SDK. For 3D head representation, we created an identical avatar but disabled other parts such as arms, body, and legs. Facial tracking and synchronization across the network are achieved by parsing 64 facial blend shape numeric parameters across clients using Photon Pun 2. Likewise, avatar body gestures, gaze movements, and object interactions are managed through similar techniques.

2.1 Emotional Cues and Reprojections

For emoji representation, we selected five emojis to depict five emotions: Neutral, Happy, Angry, Sad, and Surprise, as they are interpreted as intended, regardless of age and gender [Brants et al. 2019]. We also created four visual effects: sunshine (happiness), a fire blast (anger), dropping tears (sadness), and confetti (surprise), with color themes inspired by Plutchik's Wheel of Emotion. Unlike the 3D head, emojis and effects change based on predefined facial blend shape parameters. For vibrotactile feedback, the vibrators ($N=3$) in the VR controller held by the user's dominant hand vibrate in a heartbeat-like pattern with five beats per minute (BPM) levels representing the emotions: Neutral (75 BPM), Happiness (90 BPM), Anger (105 BPM), Sadness (60 BPM), and Surprise (120 BPM for 2 seconds followed by 75 BPM). The exaggeration of BPM rates for these emotions was decided based on a prior work [Dey et al. 2018].

The system detects users' emotions through selected facial tracking parameters and shows or hides the corresponding emojis and effects in real time. As displaying emotion-related cues near the user's gaze can enhance co-presence and emotional interdependence [Kim et al. 2020], we selected four positions where emotion cues can be attached to be near the user's gaze focus during VR interactions: **(1) User's Hand:** Attaching emotion cues to the user's dominant hand creates an embodied experience where the partner's emotions and the user's body are synergized. **(2) Partner's Hand:** Reprojecting emotions to the partner's dominant hand enables the user to stay focused on the task while still perceiving emotional cues without needing to look directly at their partner's face. **(3) User's Object:** Attaching emotions to the task object that the user is currently interacting with (or last interacted with if not currently) helps understand the partner's emotion state related to the object interacted. This can facilitate emotion consensus when both users

collaborate on the same task object. **(4) Partner's Object:** Similar to the user's object, but emotions are attached to the task object the partner interacts with. This is more passive, as it relies on the partner's interaction but still provides emotional context related to the object manipulated.

To prevent overwhelming and confusing the user with too many visual cues, we developed an Affective Panel (See Figure 1, bottom right) where users can toggle between different representations according to their preferences. This allows users to customize their experience and pick the most relevant emotional cues to their task.

3 User Experience

ExpressiveWorld provides a VR collaborative toolkit that enables collaborators to see and feel each other's emotions, which are represented in the form of visualisations via 3D head, emojis, and effects, or vibrotactile feedback in different intensities. These representations can be independently toggled on or off based on user preferences. These emotions can be attached to different positions, such as the user's or partner's hand and interacting objects. The concept offers significant benefits across various applications such as architecture, design, education, and healthcare. For example, educators can better understand and respond to students' emotional states when going over a difficult topic which leads to more effective and empathetic teaching. Likewise, remote therapy can improve patient care quality by understanding patient's emotions when providing health assessment. At SIGGRAPH Asia 2024, we will showcase an example of urban planning where visitors can experience virtual collaboration while maintaining a high awareness of their partner's emotions through customizing how emotions are represented. Concurrently, a representative acting as a local-to-the-task partner will be demonstrating and providing explanations to the conference delegates.

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